The volcano plot

Which features differ between two conditions, by how much, and how certain that is.

AT A GLANCE

Where	Statistics → Volcano
Data levels	Protein, phosphosite, phosphoprotein
Needs	Two conditions with at least two samples each; the paired test also needs a pair_id per sample.
Two sources	Volcano compares two conditions; Volcano Control compares two differences, each against its own control.
Two renderers	**Type** Interactive for exploring, Normal for a figure for print.
Fold change	$\log_2$ FC = mean(Condition 1) − mean(Condition 2). Positive means higher in Condition 1.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

CHANGED IN version 1.0.0

The fold change is computed as **Condition 1 minus Condition 2**, so a positive $\log_2$ FC means the feature is higher in the condition selected as **Condition 1**, and the axis is labelled with that subtraction in full. Earlier releases, and the volcano manual that described them, defined it the other way round — mean(Condition 2) − mean(Condition 1) — while the title of the plot read the same. Every direction in a result produced before this change is therefore reversed: what was reported as upregulated is reported

here as downregulated, at the same magnitude and the same p-value.

1 What the plot answers

Deciding that a feature has moved needs two numbers, and neither is sufficient alone: the effect size says how large the difference is and in which direction, the p-value says how well the replicates support it. The volcano puts one on each axis so that both are read at once, and the two thresholds divide the result into the classes the table reports.

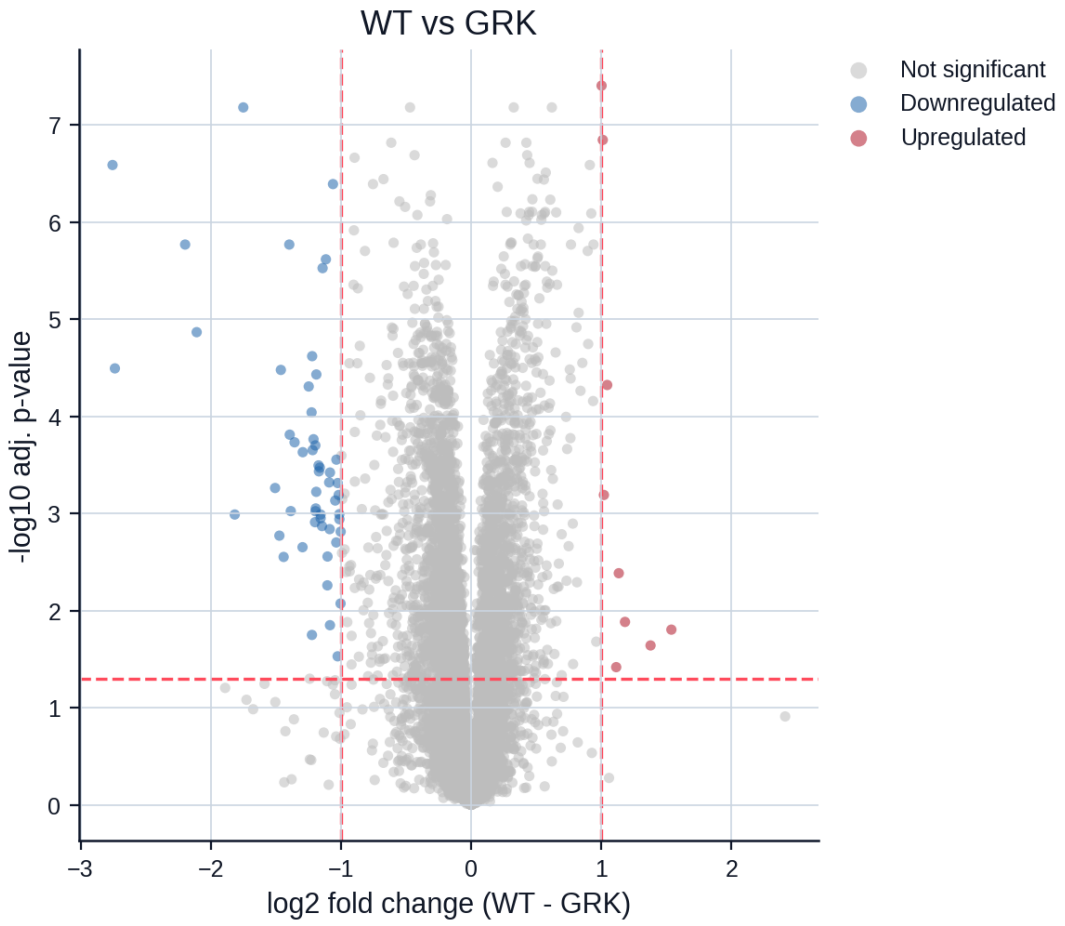


Figure 1.1. The static volcano on the example data: WT against GRK at protein level, at the default thresholds. The axis is labelled log2 fold change (WT - GRK), so the figure states its own orientation — the red points on the right are higher in WT, the blue ones on the left higher in GRK. The three dashed lines are the two $\log_2$ FC thresholds at ±1 and the p-value threshold, drawn at $-\log_{10}(0.05)$; far more features clear them on the left than on the right.

Reading it

- One point, one feature, named by **Label column**.
- The x axis, $\log_2$ FC, labelled log2 fold change (Condition 1 - Condition 2) with the real

condition names in it. Right is higher in Condition 1, left is higher in Condition 2.

- The y axis, $-\log_{10}$ of the adjusted p-value, or of the raw p-value when **Use uncorrected p-values** is on; the axis title says which. Higher is stronger evidence.
- Three threshold lines: horizontal at $-\log_{10}(\alpha)$, vertical at $\pm\tau$, for the two threshold controls.
- Colour is the class: red up, blue down, grey not significant — saturated in the interactive plot, muted in the static one.
- A feature named in **Highlight terms** keeps its class colour and gains an outline, and in the interactive plot is drawn larger and at full opacity.

A volcano need not be symmetric. An asymmetric pattern may reflect real biology, normalization assumptions, sample composition or technical effects. Check those possibilities against the design rather than adjusting the data to produce a familiar shape. Few significant points may reflect small effects, high variability, limited replication or multiple-testing correction.

For example, a log2 fold change of 1 means a twofold ratio of geometric mean intensities in the ordinary two-condition comparison; -1 means one half. An adjusted p-value of 0.01 appears at height 2. The p-value is not the probability that this protein is unchanged, and a fold-change threshold does not by itself establish biological importance. Inspect the replicate values of a candidate in the single-protein plots before interpreting it.

> EASY TO GET WRONG
>
> Do not take the direction of the x axis from memory, or from any manual written before version 1.0.0. Check it on your own data in one step: type the label of a feature you know is higher in one of the two conditions into **Highlight terms** and see which side it lands on. A feature higher in the condition you put in **Condition 1** belongs on the right; if it is on the left, the two selectors are the other way round.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Protein, phosphosite or phosphoprotein. Only the levels uploaded are offered; phosphoprotein appears once a collapse has been run.	Protein	page
Source	Volcano Control subtracts a control condition from each side first and compares the two differences; it needs four distinct conditions.	Volcano	page

Controls on this page (continued)

Control	What it does	Default	Where
Data Source	The matrix the test runs on: Data, Filtered Data or Imputed Data, whichever exist.	the most processed matrix available	page
Condition 1, Condition 2	The two conditions compared. Condition 1 is the minuend: a positive $\log_2$ FC means higher there.	first condition in the annotation	page
Condition 1 Control, Condition 2 Control	Volcano Control only. The control each side is measured against; all four conditions must differ.	first condition in the annotation	page
Type	Interactive draws the plot with hover text on every point; Normal renders a static figure for print.	Interactive	page
Width (cm), Height (cm), DPI	The size and resolution of the static figure, only while Normal is selected. The interactive plot has none: it fills its frame, Plotly's camera button saves a picture of it, and **Open Interactive Plot** opens it in a window of its own.	18 cm × 14 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF, beside **Download Plot** in the plot's header. Normal only.	PNG	page
Label column	What names a feature and is matched against **Highlight terms**: the workflow name or the gene names.	the workflow name	page
Test type	Unpaired (Student's t, equal variance), Unpaired (Welch, unequal variance) or Paired t-test (matched samples). In Volcano Control the pooled unpaired test is not offered, because the control means carry their own variance; the entries there are Unpaired (propagated variance) and Paired (propagated control variance).	Unpaired (Student's t, equal variance)	page
P-value threshold, log2 FC threshold	α and τ: a feature must be below the first and beyond the second in absolute value to be called up or down.	0.05, 1	page
Use uncorrected p-values	Applies the threshold to the raw p-value instead of the adjusted one and plots that on the y axis. It does not change the test.	off	page
Label top up-regulated, Label top down-regulated	Normal only. How many hits in each direction to name, from the top of the y axis. A label that cannot be placed clear of another is left off.	0	page
Highlight terms	Labels to emphasise, space-separated. Matching ignores case and a trailing _HUMAN; a semicolon-separated annotation matches on any token.	empty	page

Controls on this page (continued)

Control	What it does	Default	Where
Add Volcano To Report, Remove Volcano From Report	Stores the current configuration in the report list, or removes a matching entry. Plain Volcano only.	—	page

In the plot appearance panel

Control	What it does	Default	Where
Volcano points	Size, opacity, shape and colour, for all three classes at once and then per class.	6 and 0.85 significant, 5 and 0.55 not significant	appearance
Highlight size, Highlight outline	Marker size and outline colour of a highlighted feature.	10, #22c55e	appearance
Hit label size	The on-figure labels written by the **Label top** controls. Static plot only.	7	appearance
Reference line	The three threshold lines: shown or hidden, colour, dash and width. Both renderers read the same settings.	shown, dashed	appearance
Title, axis labels, font, frame, grid, background	As for every other plot; described in Part XI.	the plot's own	appearance

> **WORTH KNOWING**
>
> The appearance panel is hidden until advanced mode is on (**Ctrl** + **Shift** + **E**). Its condition-renaming card is offered here, but both renderers build the title and the x-axis label from the annotation's own condition names, so a rename reaches neither.

3 How it is computed

Choose the test from the experiment, not from which option produces more significant points. Independent biological samples and genuinely matched pairs are different designs. Welch's test permits unequal variances; the pooled test assumes equal variances. With few replicates, distributional assumptions and outlying measurements matter especially. Neither test automatically models batches, technical-replicate dependence or uncertainty introduced by imputation.

THE FOLD CHANGE

For the two selected conditions the $\log_2$-transformed values of every feature are taken from the chosen matrix, and a mean is formed within each condition over the replicates present:

$$\bar{x} = \text{mean(Condition 1)}, \quad \bar{y} = \text{mean(Condition 2)}, \quad \log_2 \text{FC} = \bar{x} - \bar{y}.$$

Positive means higher in Condition 1, negative higher in Condition 2. For ordinary $\log_2$ intensities, the difference of means is the $\log_2$ ratio of geometric means; +1 is a doubling on that processed scale. VSN is only log-like, particularly at low intensity, so its differences should not be presented as exact raw-intensity ratios.
In Volcano Control each side is centred on its own control first — every value of Condition 1 loses the mean of Condition 1 Control, and likewise for Condition 2 — before the means are taken. The axis then reads (C1 - C1 control) - (C2 - C2 control).

THE T-TEST

Unpaired, equal variance — independent samples, pooled variance. With n_1, n_2 the numbers of values actually observed for that feature and s_1^2, s_2^2 their within-condition variances:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}, \qquad SE = \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)},$$

$$t = \frac{\log_2 \text{FC}}{SE}, \quad df = n_1 + n_2 - 2, \quad p = 2\,P(T_{df} > |t|).$$

Unpaired, unequal variance pools nothing: it uses $SE = \sqrt{s_1^2/n_1 + s_2^2/n_2}$ with Welch–Satterthwaite degrees of freedom. An unpaired control-corrected comparison also propagates the two control means' variances into SE and uses Satterthwaite degrees of freedom, rather than treating estimated control means as known constants.
Paired — matched samples, read from the pair_id column of the annotation; each identifier may appear at most once per condition, and samples are ordered by it rather than by file order. The test is on the within-pair differences $d_i = x_i - y_i$, in the same orientation as everything else:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \qquad df = n - 1,$$

with $\bar{d}$ and s_d their mean and standard deviation and n the number of pairs observed on both sides, reported as n_pairs. The $\log_2$ FC on the plot is $\bar{d}$ itself, which is not the difference of the two condition means when some pairs are incomplete. With control correction, the two control-mean variance terms are added to the variance of the paired mean difference and the degrees of freedom are recomputed. This models those control means as independent components; it is not a general repeated-measures model over all four conditions.

MULTIPLE TESTING CORRECTION

One test runs per feature, so several thousand run per comparison and at $\alpha = 0.05$ several hundred features would clear the threshold on noise alone if all null hypotheses were true and tests calibrated. Under independence or appropriate positive dependence, the Benjamini–Hochberg procedure controls the false discovery rate instead — the expected share of false positives among the features called significant.

1. Sort the m p-values into ascending order, $p_{(1)} \leq p_{(2)} \leq \ldots \leq p_{(m)}$.
2. Scale each by its rank:
$$\text{adj_p}_{(i)} = \frac{m}{i} p_{(i)}.$$
3. Enforce monotonicity, walking from the largest rank down, so that an adjusted value never exceeds the one above it:
$$\text{adj_p}_{(i)} = \min\left(\text{adj_p}_{(i)}, \text{adj_p}_{(i+1)}\right),$$
and cap the result at 1.

Features whose test produced no finite p-value take no part: they are excluded from m, keep an adjusted value of NA and are classed Unavailable. m is the number of features actually tested in this comparison, not the number of rows in the dataset. adj_pval always retains the Benjamini–Hochberg adjusted value. **Use uncorrected p-values** selects the raw value for selected_pval, the y axis and classification. The correction column then says none (uncorrected) rather than Benjamini-Hochberg — so a downloaded table read without the interface still states what was done.

Classification follows from the two thresholds and the value q in selected_pval — adjusted, or raw when the uncorrected switch is on. **Upregulated**: $q < \alpha$ and $\log_2$ FC $> \tau$, so higher in Condition 1. **Downregulated**: $q < \alpha$ and $\log_2$ FC $< -\tau$, so higher in Condition 2. **Unavailable**: the row has no p-value, either because the test produced no finite one or because the row has no variation (see *Edge cases*). **Not significant**: everything else. Both conditions have to hold, which is what the two sets of threshold lines mean.

Edge cases

- A feature needs at least two observed values in each condition (for a paired test, also at least two complete pairs), or it is not tested. It is not drawn on the plot and is not in the result table, so a feature measured in only one condition is never shown at an extreme fold change. It is not lost silently either: a warning above the plot reads *u* of *T* rows were not tested: fewer than 2 values in a compared condition or, for paired tests, fewer than 2 complete pairs., and the rows are listed in their own table, described below.
- A condition with a single replicate makes every feature fail that rule and the comparison is refused. The paired test additionally needs two complete pairs.
- A constant condition is flagged in zero_variance. One constant group does not necessarily make the whole standard error zero: variation in the other group still contributes. A row without any variation — every compared group constant; for a paired test, identical

differences; for Volcano Control, the controls constant too — has a standard error of zero and an undefined test. It gets no p-value, is classed Unavailable and counted under No p-value, and a warning on the page says how many rows this applies to. Inspect rounding, imputation and detection floors before interpreting these cases.

- A p-value at or below 10^{-300} is clipped there before the logarithm, so the y axis has a ceiling near 300.
- Imputed Data tests imputed values as though measured.

4 The table below the plot

The table is the plot as numbers: one row per feature tested, in the order of the matrix, carrying the label column, every sample column of the two conditions as it went into the test, and the following.

Columns in this table

Column	What it holds
the label column	The feature, as **Label column** selects it. Phosphoprotein data also carries site_num and PTM_Collapse_keys where they exist.
log2FC	mean(Condition 1) − mean(Condition 2), or the mean of the paired differences.
pval	The raw two-sided p-value of the t-test.
adj_pval	The Benjamini–Hochberg adjusted p-value, retained even when the display uses raw p-values.
selected_pval, correction	The p-value used for classification and plotting, and whether the selected basis is adjusted or raw.
neg_log10_selected_pval	$-\log_{10}$ of the value on the y axis, clipped at 10^{-300}.
significance	Upregulated, Downregulated, Not significant or Unavailable.
zero_variance	True where every observed replicate of one condition holds the same value (for a paired test, where the differences are identical). Where every compared group is constant the row has no p-value; otherwise the p-value is unaffected and the flag says where it came from.
n_pairs	Paired test only: complete pairs behind the row.

The counts above the table — Rows, Upregulated, Downregulated, Not Significant and No p-value — are over the same set of tested features.

When some rows could not be tested, a second table, Rows not tested, follows the result table with its own **Download CSV** (the file name ends in _not_tested.csv). It has one row per untested feature: its identifier columns, n_<condition> and mean_<condition> for each compared condition — the observed count and mean before any control correction — n_pairs

for a paired test, and reason, for example fewer than 2 values in GRK. These are often the on/off cases worth inspecting by hand.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
volcano_data = volcano_data_f(data, meta, "WT", "GRK", workflow="Protein")
fig = volcano_plot(volcano_data, "WT", "GRK")
```

R

```
volcano_data <- volcano_data_f(data, meta, "WT", "GRK", workflow = "Protein")
fig <- volcano_plot(volcano_data, "WT", "GRK")
```

The volcano is two calls and not one: volcano_data_f runs the tests and returns the table — fold changes, p-values, adjusted p-values and classes — and volcano_plot draws that table, so the numbers can be inspected, corrected or exported before anything is plotted. Give both calls the same in_pval and in_log2fc: the plot colours its points from the significance column the table already carries, but draws its three threshold lines from its own arguments, so a mismatch leaves the lines somewhere the colours do not agree with. The package draws the figure with Plotly, so what comes back is the *interactive* volcano with hover text on every point; the static figure above is the application's own second renderer, which the packages do not carry.

6 Exporting

Type chooses between two renderings of identical numbers. Interactive keeps its gridlines and its full frame and names a point on hover, which is how a feature is identified while exploring; Open Plot opens it in its own tab. Normal is the version for a paper: no grid, two spines, the muted palette, the legend outside the plotting area, the points rasterised, and nothing labelled unless the **Label top** controls ask for it; Open Image gives the PNG. The table downloads separately as CSV, named after the two conditions.

The static figure is rendered at a fixed 18 by 14 cm and 300 dpi. Unlike the quality-control plots this page offers no width, height or resolution controls, which is deliberate: a volcano

is read at one size, and the two threshold lines and the label placement are laid out for it.

7 Further reading

SciPy: independent-sample t-tests describes the equal-variance and Welch tests. SciPy: false discovery control explains adjusted p-values and the assumptions behind Benjamini–Hochberg control. The application may pin a different library version; these links are conceptual references.

Behind this page

Function	What it produces
volcano_data_f	The table: fold changes, p-values, adjusted p-values and classes
volcano_data_f_control	The same, with each side centred on its own control condition
volcano_plot	The interactive volcano
volcano_plot_h	The interactive volcano with named features highlighted